

Unit 13, Lesson 11: Calculating and Interpreting Average Rate of Change



Benchmark

MA.912.F.1.3 Calculate and interpret the average rate of change of a real-world situation represented graphically, algebraically, or in a table over a specified interval.



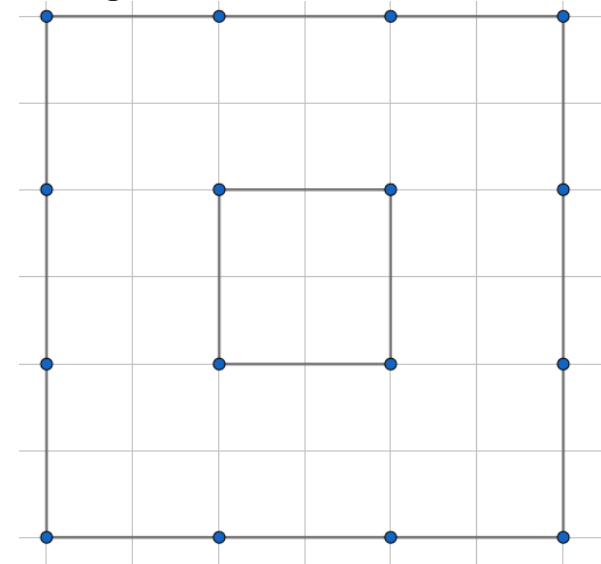
Learning Targets

- ☐ I can calculate the average rate of change of a real-world situation represented graphically, algebraically, or in a table over a specified interval.
- ☐ I can interpret the average rate of change in a real-world context.



13.11.1 Warm-Up: Critical Thinking

- Two squares are shown on the grid that were created using line segments.



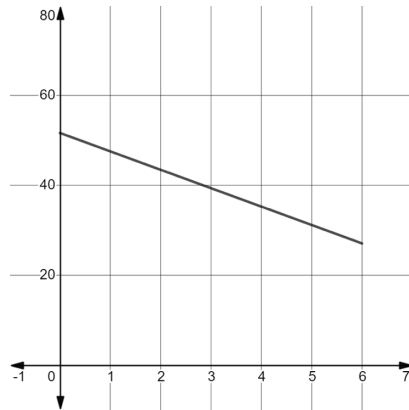
Move 4 line segments to create 3 squares.



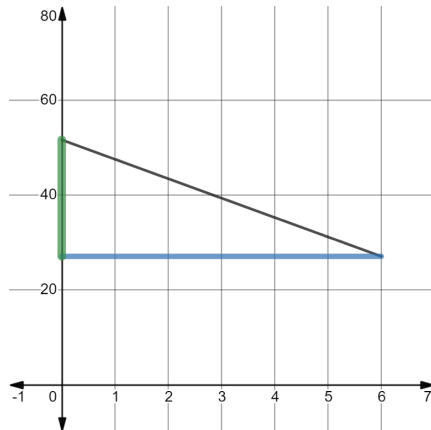
13.11.2 Guided Instruction: Average Rate of Change

1. In 2005, the percentage of the population of the Democratic Republic of the Congo living at or below the International Poverty Line was 51.6. The table and the graph show the percentage of the population from 2005 to 2011, where x is the number of years since 2005 and y is the percentage of the population living at or below poverty level.

Congo 2005 – 2011	
x	y
0	51.6
1	47.5
2	43.4
3	39.3
4	35.2
5	31.1
6	27



- a. Using the graph of the relationship, discuss with your partner how the triangle formed shows the slope.

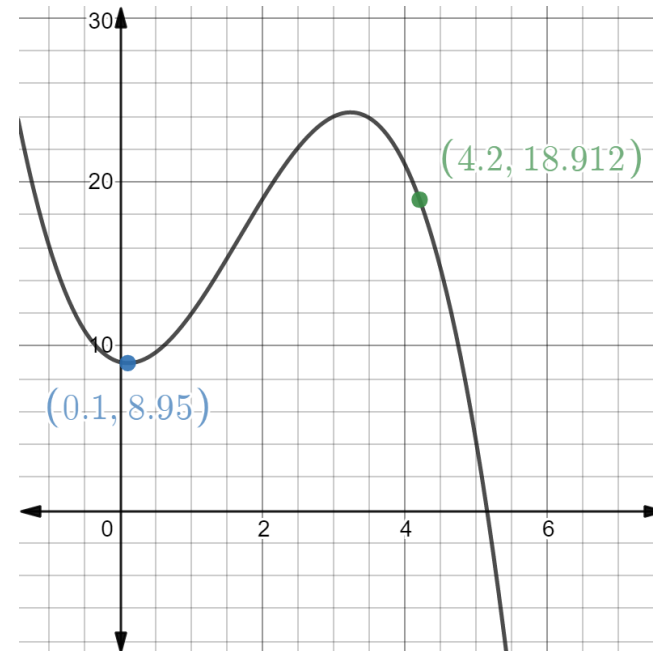


- b. Complete the table to find the slope.

Green Line Segment	$\Delta y =$	$\frac{\Delta y}{\Delta x} =$
Blue Line Segment	$\Delta x =$	

- c. Interpret the slope in terms of the context.

2. The graph of a function is shown. Points $(0.1, 8.95)$ and $(4.2, 18.912)$ are on the function.



- a. Draw a line that connects the points $(0.1, 8.95)$ and $(4.2, 18.912)$.

The line drawn is a secant line.

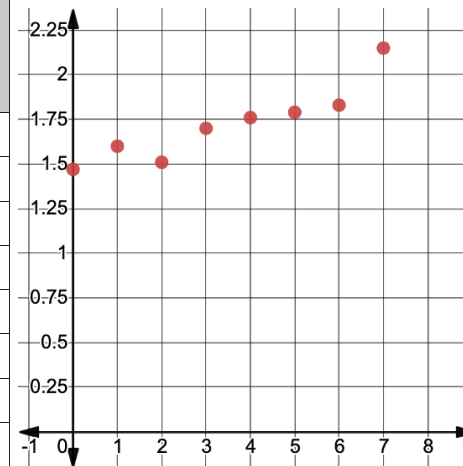
A **secant line** is a line that passes through two points of a curve.

- b. Find the slope of the secant line that passes through the points $(0.1, 8.95)$ and $(4.2, 18.912)$.

The **average rate of change** is the slope of the secant line. It is the change in the y -values divided by the change in the x -values for two distinct points.

3. The average price for 16 ounces of creamy peanut butter for the years 1984 to 1991 is shown in the table. The values are plotted on the coordinate grid as shown.

Years Since 1984	Average Price of Peanut Butter (\$ per pound)
0	1.471
1	1.601
2	1.514
3	1.695
4	1.761
5	1.792
6	1.834
7	2.147



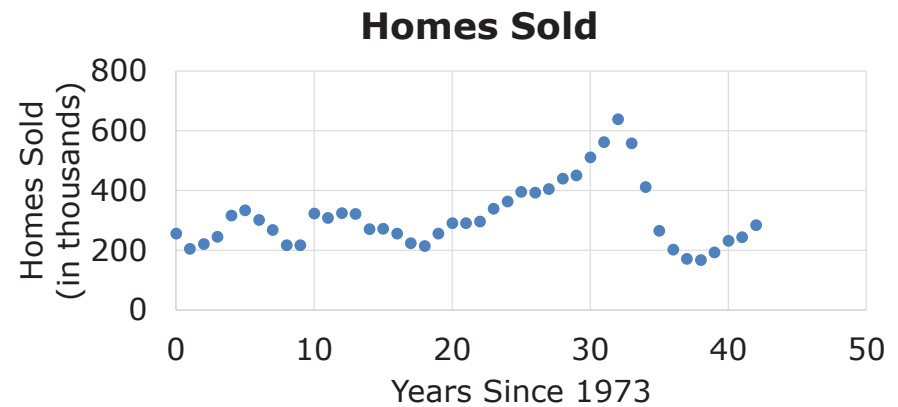
- a. Find the average rate of change for the interval 1985 to 1989. Include units.
- b. Find the average rate of change for the interval 1984 to 1991.

- c. Interpret the average rate of change for the interval 1984 to 1991 in terms of the context.

- d. What do you notice about the average rates of change for 1985 to 1989 and 1984 to 1991?

4. Mathematicians and economists often want to evaluate changes that have occurred over different time intervals to analyze trends and make predictions.

The graph shows the number of homes sold in the U.S., in thousands, from 1973 to 2015.



- a. Draw a line from the first data point to the last data point.
- b. Discuss with your partner why finding a rate of change using the first data point to the last data point would not give much information about how the number of sales has changed.

The table shows some of the values from the graph.

Years Since 1973	Homes Sold (in thousands)	Years Since 1973	Homes Sold (in thousands)
20	292	32	639
21	292	33	559
22	297	34	412
23	340	35	266
24	364	36	203
25	396	37	172
26	394	38	168
27	406	39	194
28	440	40	232
29	451	41	244
30	511	42	285
31	562		

- c. Find the average rate of change for the time period of 1993 to 2005.
- d. Interpret the average rate of change for the interval 1993 to 2005 in terms of the context.

5. The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, U.S. passengers traveled on a train for the period of 1970 to 2000, where x is the number of years since 1970.
 - a. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period from 1986 to 2000?
 - b. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period from 1970 to 1980?
 - c. Discuss with your partner what the slope indicates about whether the graph is increasing or decreasing on the given interval.



13.11.3 Try It: Average Rate of Change

1. Mrs. Wilson's class conducts an investigation to find out how quickly the height of water in a half-gallon jug changes after a small hole is created near the bottom of the jug. The class finds the function $H(x) = 0.01x^2 - 2.77x + 136.51$ models the height, in millimeters, of water x seconds after the hole is created.

What is the average rate of change, in millimeters per second, for the period of 37 to 53 seconds? Round your answer to the nearest hundredth.

2. The table shows the number of tons of blueberries that were imported to the US from 2005 to 2013.

Year	Blueberries Imported (tons)
2005	26,319
2006	32,602
2007	35,125
2008	52,111
2009	60,450
2010	76,743
2011	87,336
2012	96,367
2013	103,502

- a. Determine the average rate of change from 2007 to 2012.
- b. Interpret the average rate of change for the interval 2007 to 2012 in terms of the context.



13.11.4 Wrap-Up: Ticket Out the Door

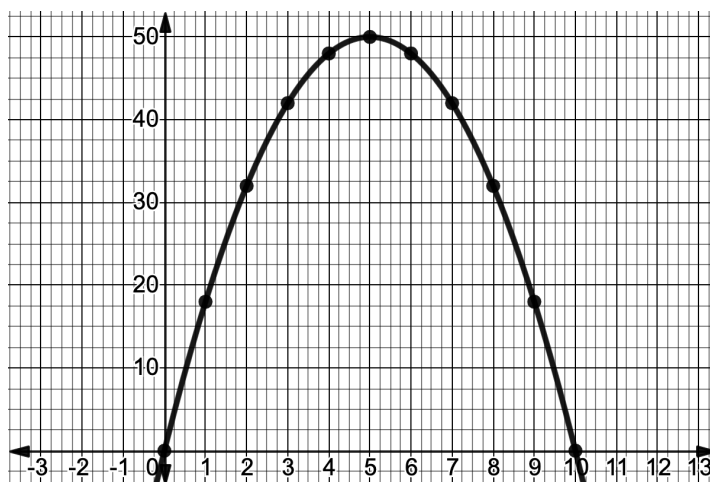
A rain gutter is made from a 20 inch (in.) aluminum sheet with the edges folded up, as shown.

Flat Aluminum Sheet	Sheet Folded for Rain Gutter
 20 in.	 (20-2x) in.

The function $A(x) = x(20 - 2x)$ models the cross-sectional

area of a rain gutter made from the sheet, where x is the height of the rain gutter.

The graph of the function is shown.



1. What is the average rate of change for the interval of heights of 1 in. to 5 in.?
2. Interpret the average rate of change for the interval of 1 in. to 5 in. in terms of the context.

Unit 13, Lesson H2: Combining Functions



Benchmark

MA.912.F.3.1 Given a mathematical or real-world context, combine two functions, limited to linear and quadratic, using arithmetic operations. When appropriate, include domain restrictions for the new function.



Learning Targets

- ☐ I can combine two functions using arithmetic operations.
- ☐ I can determine domain restrictions for a real-world context where two functions are combined.